Experimental Design & ANOVA

Department of Statistical Sciences, University of Cape Town

What you have done so far

Regression analysis:

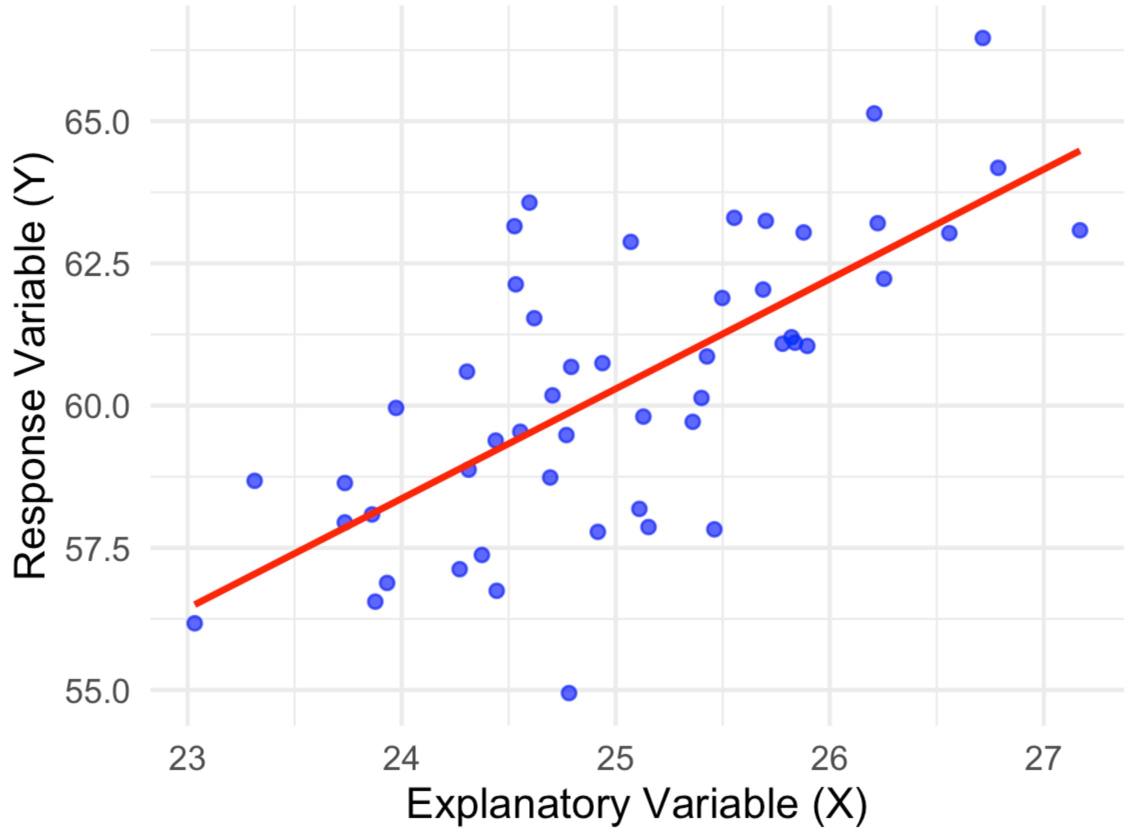
- Understanding the relationship between a response variable and explanatory variables.
- Prediction of future response value for given values of explanatory variables.
- Built a linear model that explains the response variable as a function of the explanatory variables.

Regression Model

We describe a response variable Y as a linear function of an explanatory variable X:

$$Y_i = \beta_0 + \beta_1 X_i + e_i, \quad e_i \sim N(0, \sigma^2)$$

- β_0: intercept
- β_1: slope (change in Y for a one-unit change in X)
- e_i: random error term



Examples

In the field of business science:

- Commodity tarders: forecast future prices of commodities based on past prices and other factors.
- Economists: predict GDP growth based on interest rates, inflation, and other economic indicators.
- Real estate agents: predict property values based on location, size, and amenities.
- Marketing analysts: predict customer spending based on demographics and past purchasing behavior.

But first we must collect data!

- Two fundamental ways: **observation** and **experimentation**.
- **Observational Study:**
    - No intervention
    - Measurements taken as they occur in nature.
    - Values of independent variables are **not controlled** (set in advance)

Example

- Understand the relationship between the price of a house and a set of features (e.g., size, number of bedrooms, location).
- Collect data on houses that have already been sold, without manipulating any variables.
- Values of the explanatory variables have not been set in advance, we simply recorded them.

Price	Bedrooms	Location	Size
320949.1	4	Urban	133.1857
316215.2	1	Urban	143.0947
260923.2	3	Urban	196.7612
260568.9	4	Urban	152.1153
274890.1	3	Rural	153.8786
374803.0	5	Urban	201.4519

- Vasts amounts of information that we generate
- Mobiles, computers, smart watches, IoT, sattelites, traffic lights etc
- Mostly observational data
- Not measured under manipulated or controlled conditions, just recorded as it happens

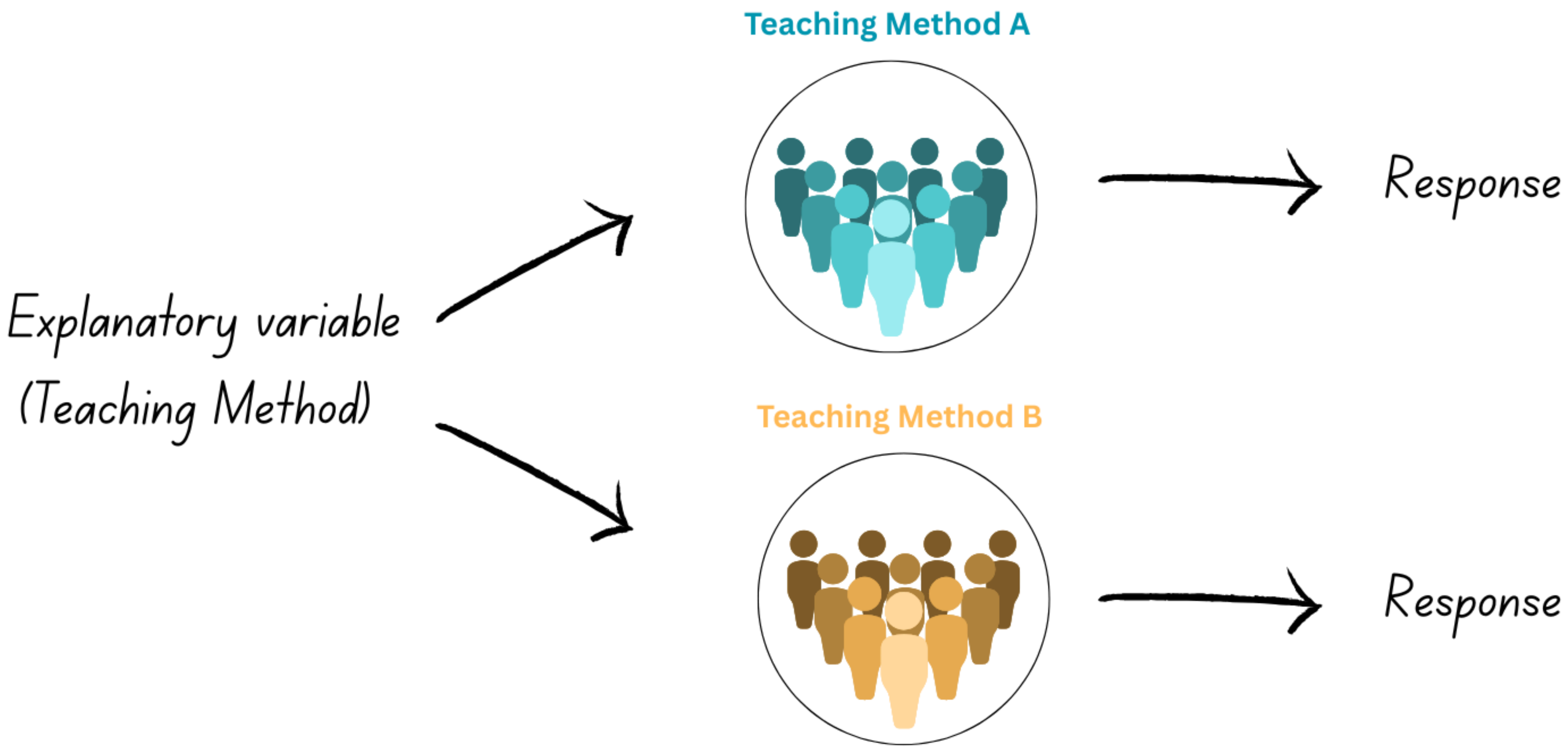
Observational vs Experimental Studies

Experimental Study:

- Researchers actively **manipulate** conditions to investigate specific questions
- Values of independent variables are controlled (set in advance)

Example

- Suppose we want to compare student performance under two teaching methods (explanatory variable).
- **Randomly** assign students to two groups: one group uses a new method, and the other uses a traditional method.
- Measure the performance of both groups to see if there is a significant difference.
- **Manipulated** the conditions under which we measure the response variable (student performance) by setting the explanatory variable (teaching method) in advance.


Teaching Method A
Teaching Method B
Explanatory variable
(Teaching Method)
Response
Response

Observational vs Experimental Studies

For each of these identify the response and explanatory variable and whether it is an observational or expeirmental study:

1. Researchers surveyed 1000 people about their exercise regime and followed them for 10 years to assess Type 1 diabetes risk.
2. Researchers recorded the daily coffee intake and heart rates of 500 adults over a month.
3. A scientist applied different fertilizers to plots of wheat to measure their effect on crop yield.
4. Medical records were analyzed to determine if patients with high cholesterol were more likely to have strokes.
5. Students were randomly assigned to study with music or in silence to test its effect on test performance.

Observational vs Experimental Studies

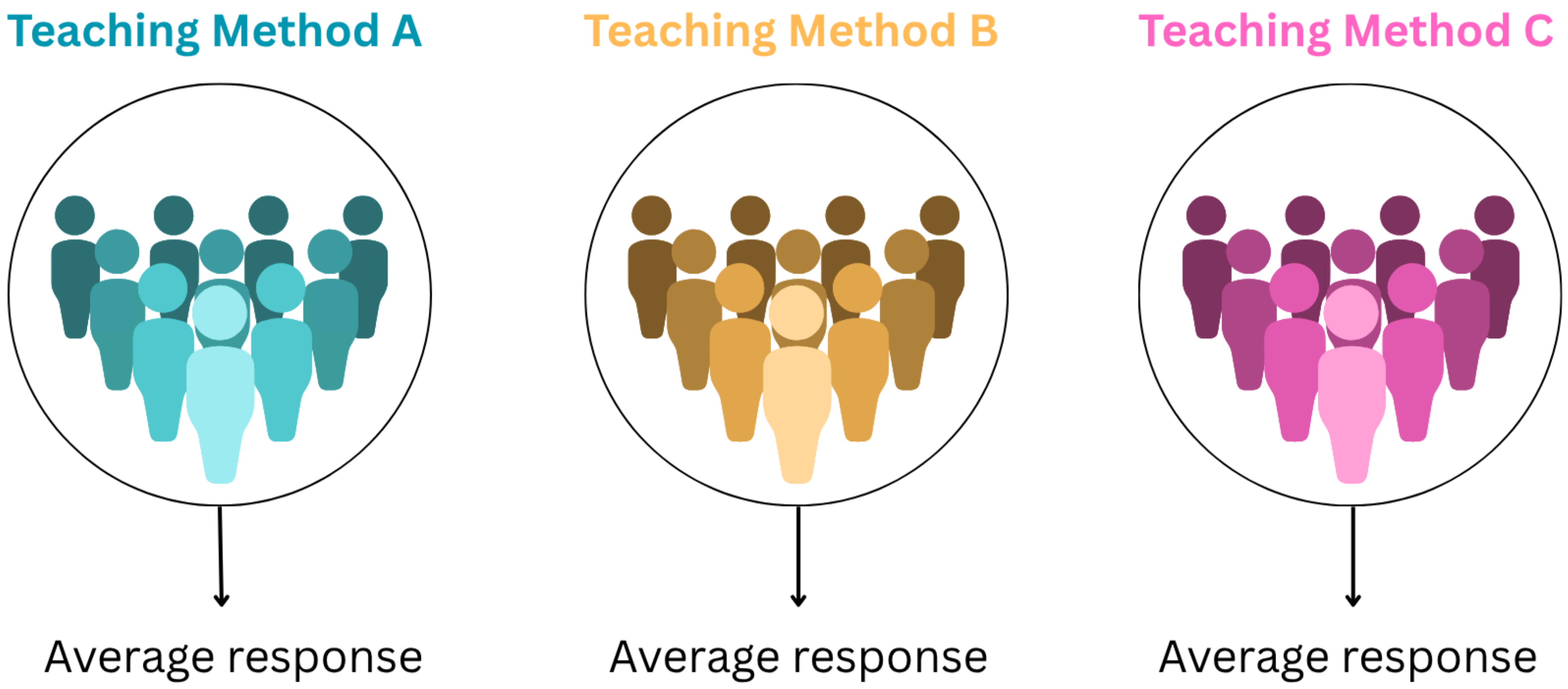
- Consider the first example about exercise and diabetes
- Researchers then found that people who exercised more had a lower risk of diabetes.
- Could we confidently conclude that more exercise = lower risk of diabetes?
- What other factors could influence diabetes risk?
- Now what if we randomly assigned people to two different exercise regimes?

Observational vs Experimental Studies

- Experiments = cause-and-effect relationships
- Control and holds constant (as best we can) all other factors that might affect the response
- Observational studies = third / confounding variable problem
- Confounding variable is related to both the response and explanatory variables
- making it difficult to determine the true relationship between them.
- Correlation ≠ causation
- A statistically signficant relationship between a response and predictor variable DOES NOT imply a cause-and-effect relationship.

Goal

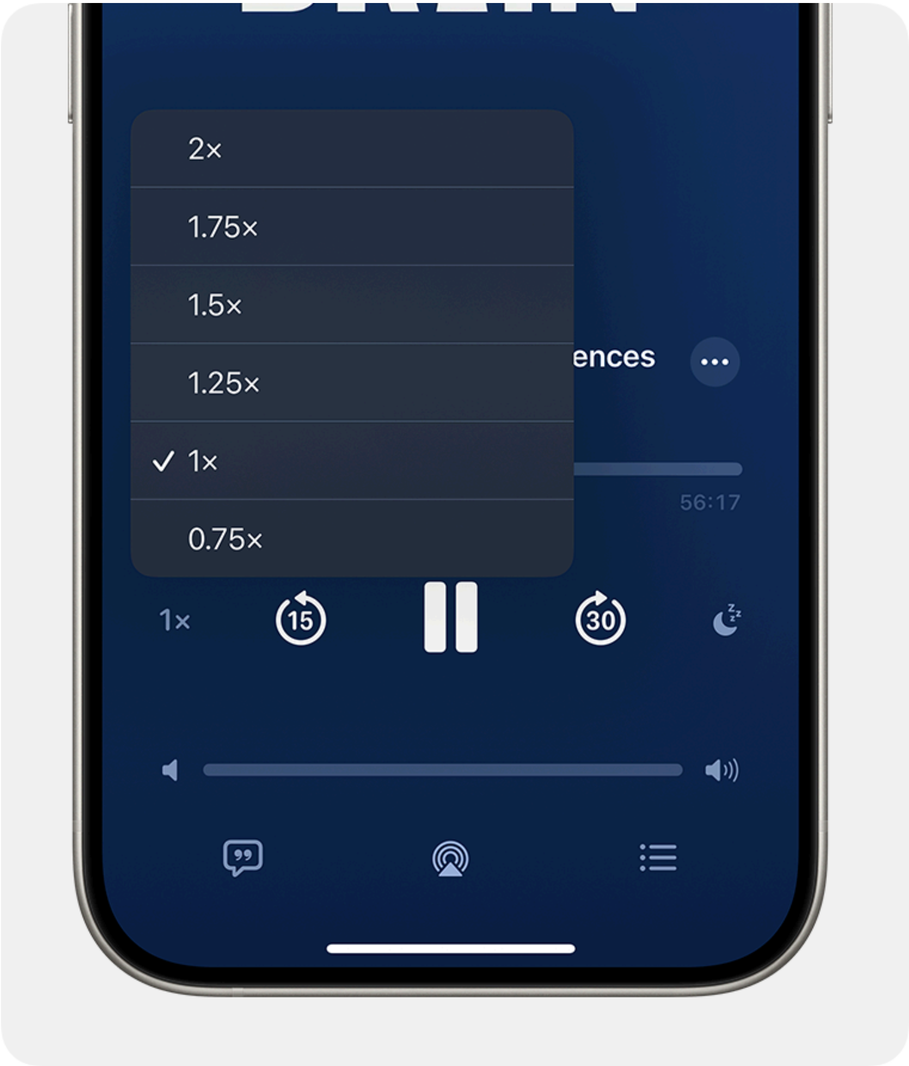
Analyse data from comparative experiments where the aim is to compare the mean response for different settings/levels/categories/groups of explanatory variables.



Working example

Question: Does playback speed affect comprehension?

A study done by Chen et al. (2024) conducted an experiment to answer this question!



Working example

- **Objective**: Investigate whether increasing playback speed affects comprehension of audio-only content.
- **Method**:
  - 180 undergraduate participants listened to two podcasts at either **1x, 1.5x, 2x, or 2.5x speed**.
  - They were randomly assigned one of the four playback speeds.
  - Participants completed multiple-choice comprehension tests after listening.
  - Average score taken as final outcome.

Some terminology

- **Response Variable:** The outcome measured in the experiment.
- **Factor:** Variable being manipulated (i.e. explanatory variables in regression)
- **Factor levels:** Different levels/settings of the factor set in advance.
- **Treatments:** Levels or combination of treatment levels (if more than one factor being studied) being compared.
- **Experimental Units:** Subjects or objects to which treatments are applied
- **Observational Units:** The entities on which the response is measured (often the same as experimental units, but not always).
- **Replicates:** number of experimental units per treatment.

Example

1. A lecturer compares the effect of teaching method on student performance. Three groups of students are taught the same material using either lectures, online videos, or group discussions. Their test scores are then measured.
2. A store manager investigates how shelf placement affects sales. The same brand of cereal is placed on either the bottom, middle, or top shelf for a week in a random order and then this is repeated for an additional three weeks. Weekly sales are recorded for each shelf placement.

a. Response - Ask: what variable am I measuring?

b. Factor(s) - Ask: what variable(s) have been set to certain levels/categories in advance?

c. Factor levels - Ask: what are the different levels of the factor(s)?

d. Treatments - Ask: what are the different combinations of factor levels?

e. Experimental units - Ask: what are the entities/things to which the treatments are applied?

f. Observational units - Ask: what are the entities on which the response is measured?

Designing experiments

- A good experiment is one that allows us to detect differences in the response variable between treatments (if there is one).
- Think of this as ensuring that the receipt of a signal (differences in means) is not drowned out by any background noise (variability in the response variable).
- Can do this by:

1. Enhancing the signal
2. Decreasing the noise

Designing experiments

- Standard error of most estimators:
  - proportional to σ (a measure of data variation or noise)
  - inversely proportional to sample size (a measure of volume of the signal)
- Consider: sample mean as an estimator of the population mean
- Smaller standard errors = more precise estimates and greater confidence in that estimate!
- Q: Based on this formula, what is one way in which we can make the standard error smaller?

The three R's of experimental design

1. Randomisation
2. Replication
3. Reduction of experimental error variance

Randomisation

The assignment of treatments to experimental units **at random**.

- Random does not mean haphazard/uncontrolled/unplanned.
- Random means with equal probability.
- Different levels of playback speed (i.e. treatment) was randomly assigned to each student.
- Consider the first 10 students (i.e. experimental units):

Exp unit	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**…**
Treatment	2x	2x	2x	1.5x	2x	1.5x	2x	1.5x	1.5x	1.5x	…

Randomisation in R

```
set.seed(123)

treats <- c("1x","1.5x","2x","2.5x")
exp_units <- 1:180
random <- sample(treats, 180, replace = TRUE)
```

```
      exp_units random
 [1,] "1"       "2x"
 [2,] "2"       "2x"
 [3,] "3"       "2x"
 [4,] "4"       "1.5x"
 [5,] "5"       "2x"
 [6,] "6"       "1.5x"
 [7,] "7"       "1.5x"
 [8,] "8"       "1.5x"
 [9,] "9"       "2x"
[10,] "10"      "1x"
```

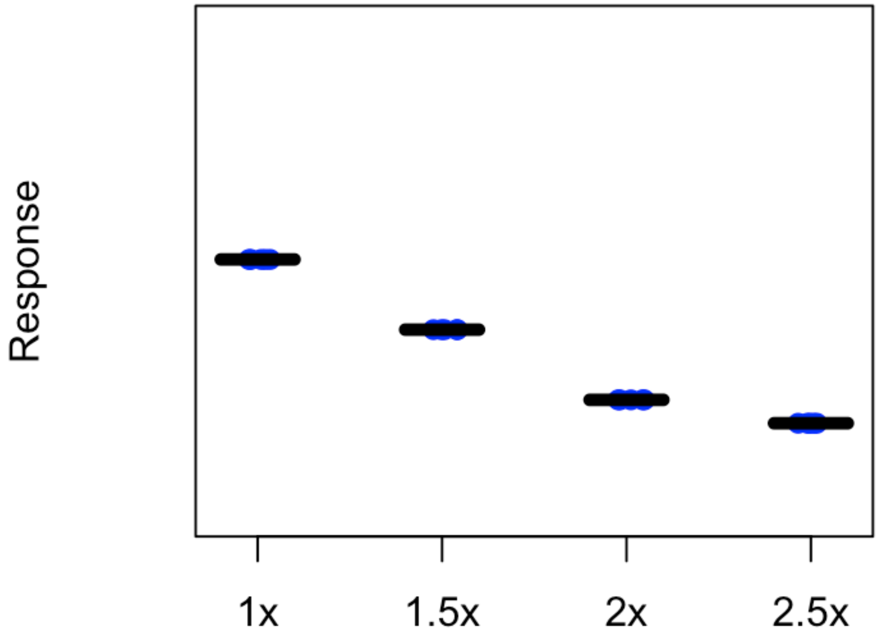
Why randomise? Enhance the signal 🔊

What would happen if we didn’t randomise?

- Any difference between the treatments could be due to differences in experimental units!
- Average out systematic effects (anticipated or not), extraneous factors not directly controlled (prevent confounding).
- Separate treatment effect from group/experimental unit effect.

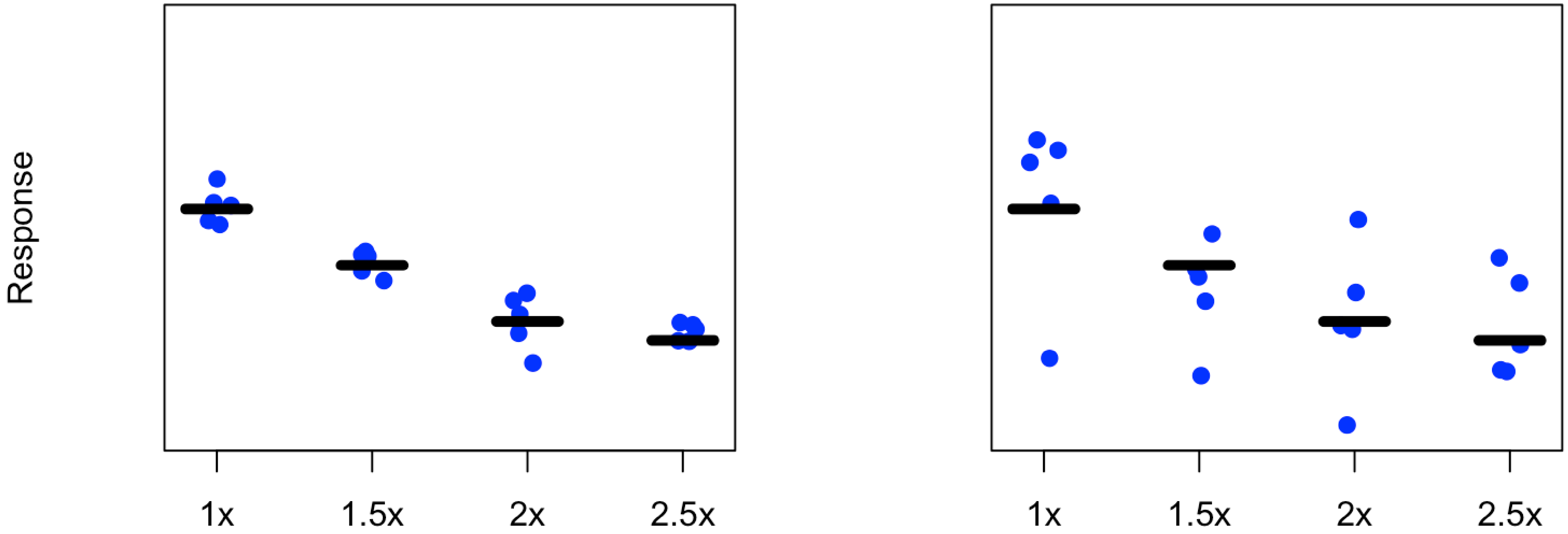
Replication

- When a treatment is applied **independently** to more than one experimental unit = replicated.
- If all experimental units were **exactly** identical, it would be easy to establish cause-and-effect relationships.



All things vary!

Experimental error variance: differences among experimental units within the same treatment



Why replication? Turn up the volume! 🔊

- By replicating we can quantify the variability of the response variable within each treatment group (exp error variance)
- and compare it to the variability between groups.
- Increases the signal (more data)
- Quantify the noise.

Reduce experimental error variance (noise)

Decrease the noise by:

1. Controlling conditions of experiment, choosing similar experimental units and randomisation.
2. Blocking if there are clear differences between groups of experimental units that might influence the response.

Blocking: Reduce the noise 🎧

Blocking a technique where similar experimental units are grouped into blocks.

- Any nuisance variables that might influence the response.
- Account for differences between blocks to isolate differences between treatments.

Example 1

- Imagine a hypothetical shop
- Suppose we want to test three strategies for stocking inventory (A, B and C)
- For the next 6 days, we use the three strategies in a random order:

Day	**1**	**2**	**3**	**4**	**5**	**6**
Strategy	A	B	C	A	B	C

- But what about day to day variability?
- Wouldn't be able to separate treatment effects from day-to-day variability = confounded effects.
- A better strategy would be to apply each strategy each day!
- Ensures that each strategy is tested under similar conditions.
- So we block by day and apply each strategy randomly at different times:
- Day 1: A, C, B
- Day 2: B, C, A etc.

Designing an experiment

- What are the treatment factors and treatments?
- What is the outcome/response?
- Experimental units?
- Any blocking factors?
- How many replicates?
- How will randomisation be applied?

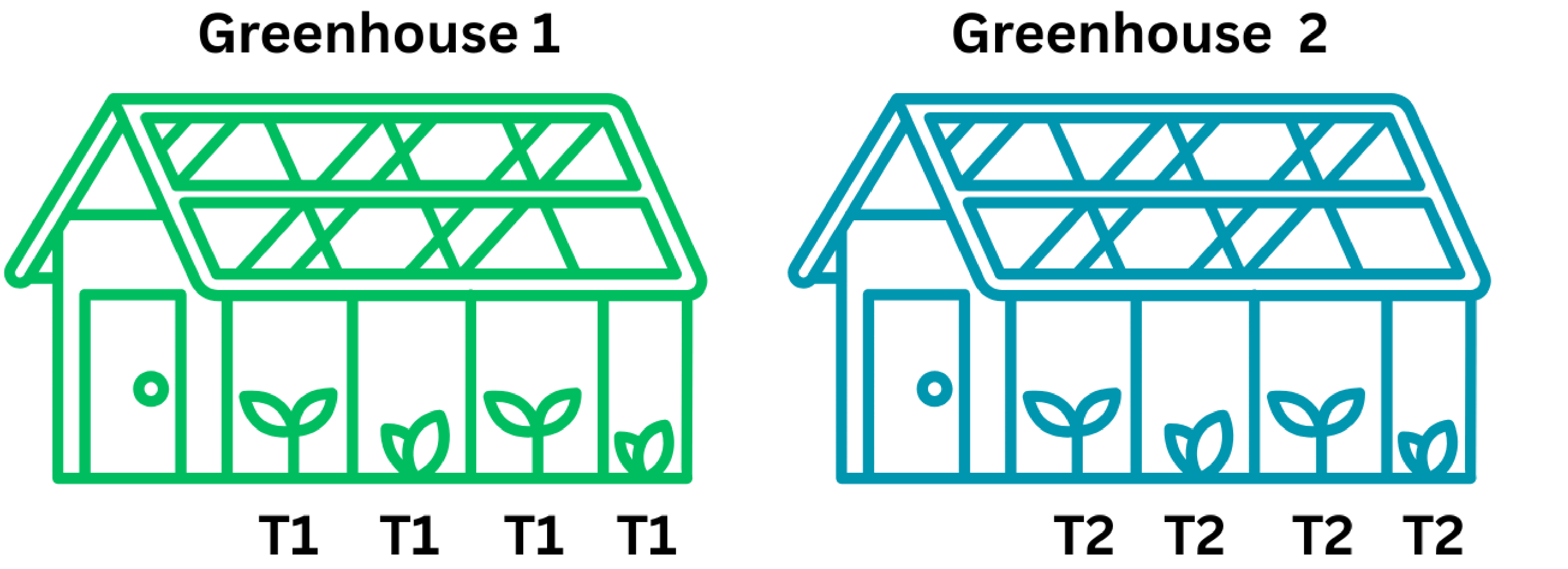
Need to choose *treatment* and *blocking structure*.

Treatment structure

- Single-factor:
  - Treatments are the different levels of one factor (e.g., playback speed: 1x, 1.5x, 2x, 2.5x).
- Factorial:
  - More than one factor is tested simultaneously.
  - Treatments are combinations of factor levels (e.g., playback speed × noise level).

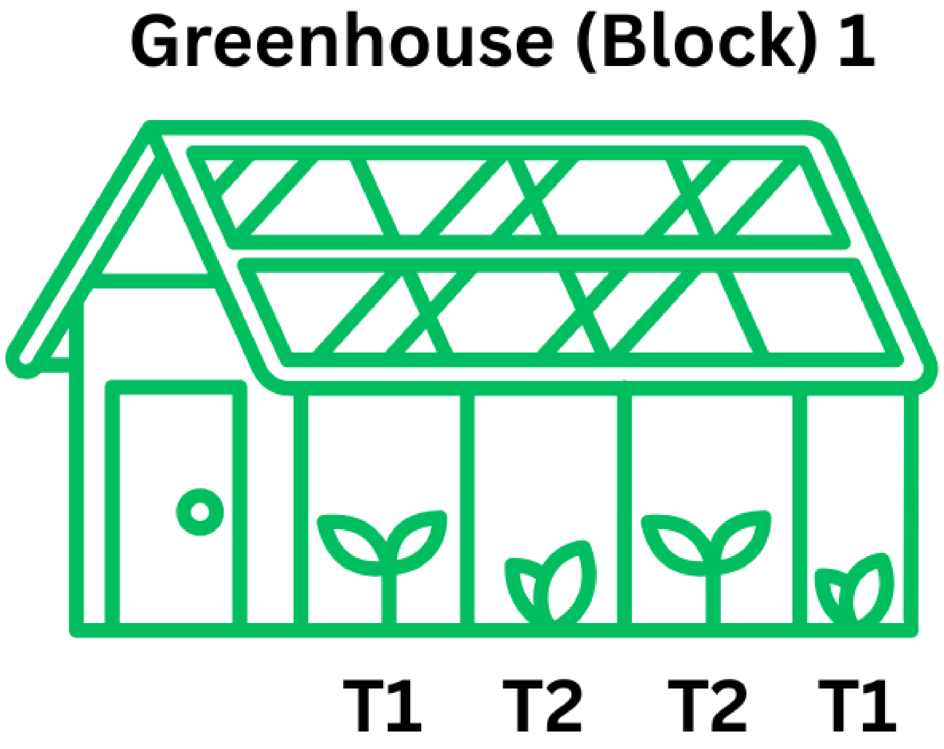
Blocking structure

- Determined by available experimental units or conditions of experimental setup.
- Example:
  - We want to test the effect of temperature on plant growth.
  - We have two greenhouses.



Blocking structure

- Block by greenhouse and randomly assign plants to different temperature treatments within each greenhouse.



Designs we cover:

1. Completely Randomized Design (CRD)

- When all experimental units are homogeneous.
- Treatments are randomly assigned to all experimental units.
- Example: Randomly assigning patients to Drug A or Drug B without any further grouping.

2. Randomized Block Design (RBD)

- When experimental units are not homogeneous (blocking required).
- Treatments are randomized within each block.
- Example: Dividing students into blocks by proficiency level before testing different teaching methods.

Design dictates analysis!

The type of design determines the statistical method:

- SF CRD → One-way ANOVA.
- SF Randomized Complete Block Design → Two-way ANOVA without interaction.
- Factorial treatment structure (CRD or RCBD) → Two-way or multi-way ANOVA with interaction(s) between treatment factors.

Single Factor CRD

- Simplest design
- Factor A with a levels (e.g., playback speed: 1x, 1.5x, 2x, 2.5x).
- The a treatments are randomly assigned to r experimental units.
- This means: $N = a \times r$ total observations.
- Playback example: $a = 4, r = 45$ and $N = r \times a = 45 \times 4 = 180$
- Response variable Y_{ij} for each treatment i form 1 to a and replicate j from 1 to r.

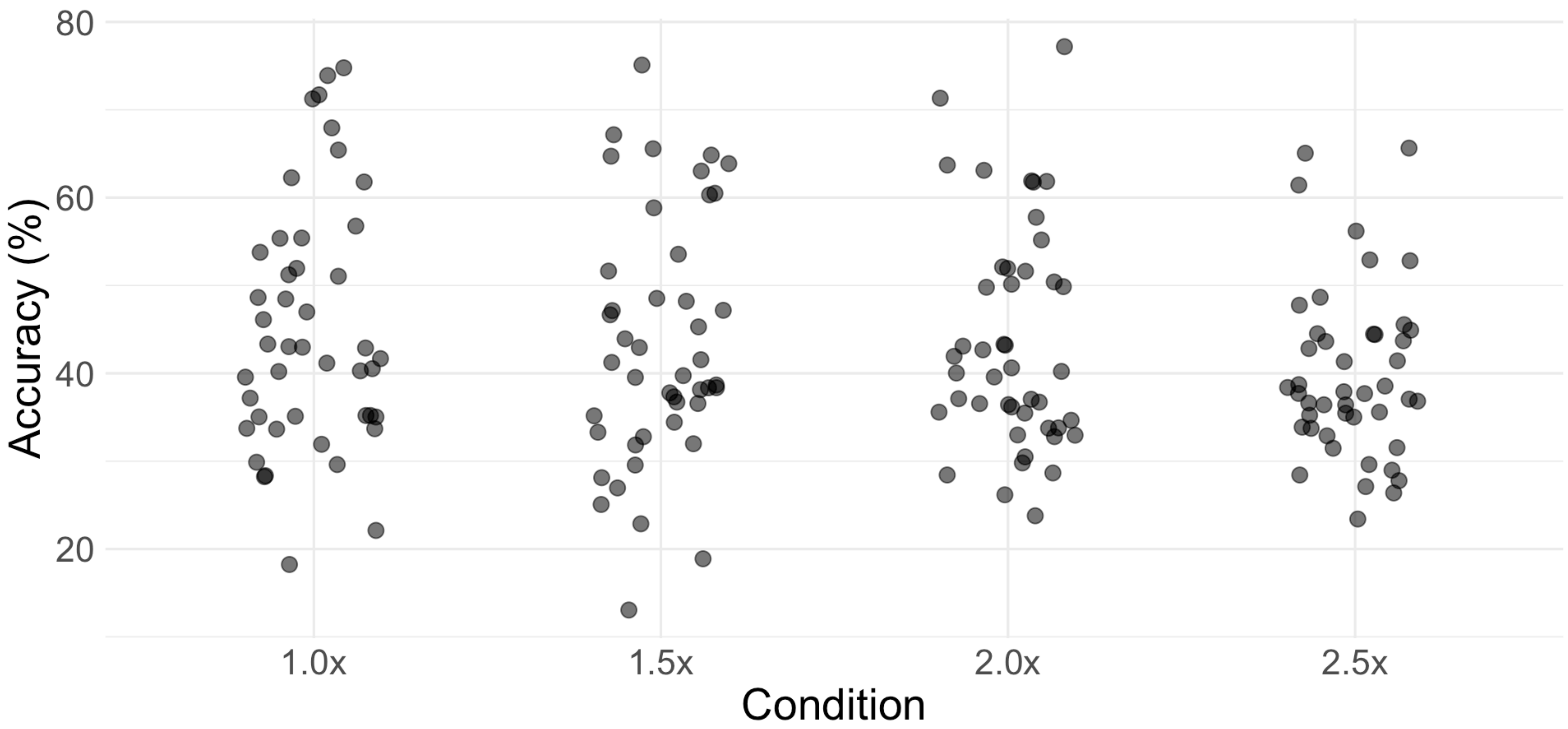
Playback data

```r
playback <- read.csv("Data/playback.csv")
head(playback)
```

```
  Participant.ID Condition Comprehension
1     76b0823c2f      2.0x         33.33
2     24ae54af6f      2.0x         50.00
3     3add152a7f      1.5x         65.00
4     e7d55aa954      2.0x         36.67
5     f4f7a8b549      1.0x         53.33
6     3ad6d70ab2      1.0x         41.67
```

Goal: Build a statistical model to compare treatment means.

Visualisation of sample data



Building a Statistical Model

- Regression: $Y_i = \beta_0 + \beta_1 X_i + e_i$
- Our goal is to compare the mean response between groups/treatments
- What do you think this model will look like?

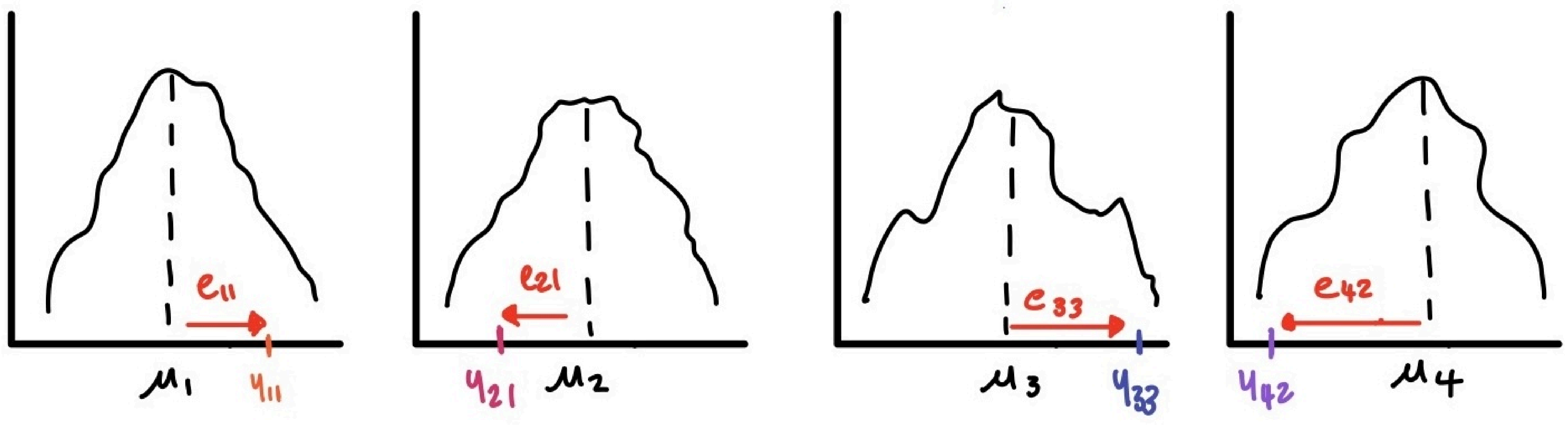
ANOVA Model

The simplest model we can hypothesise is:

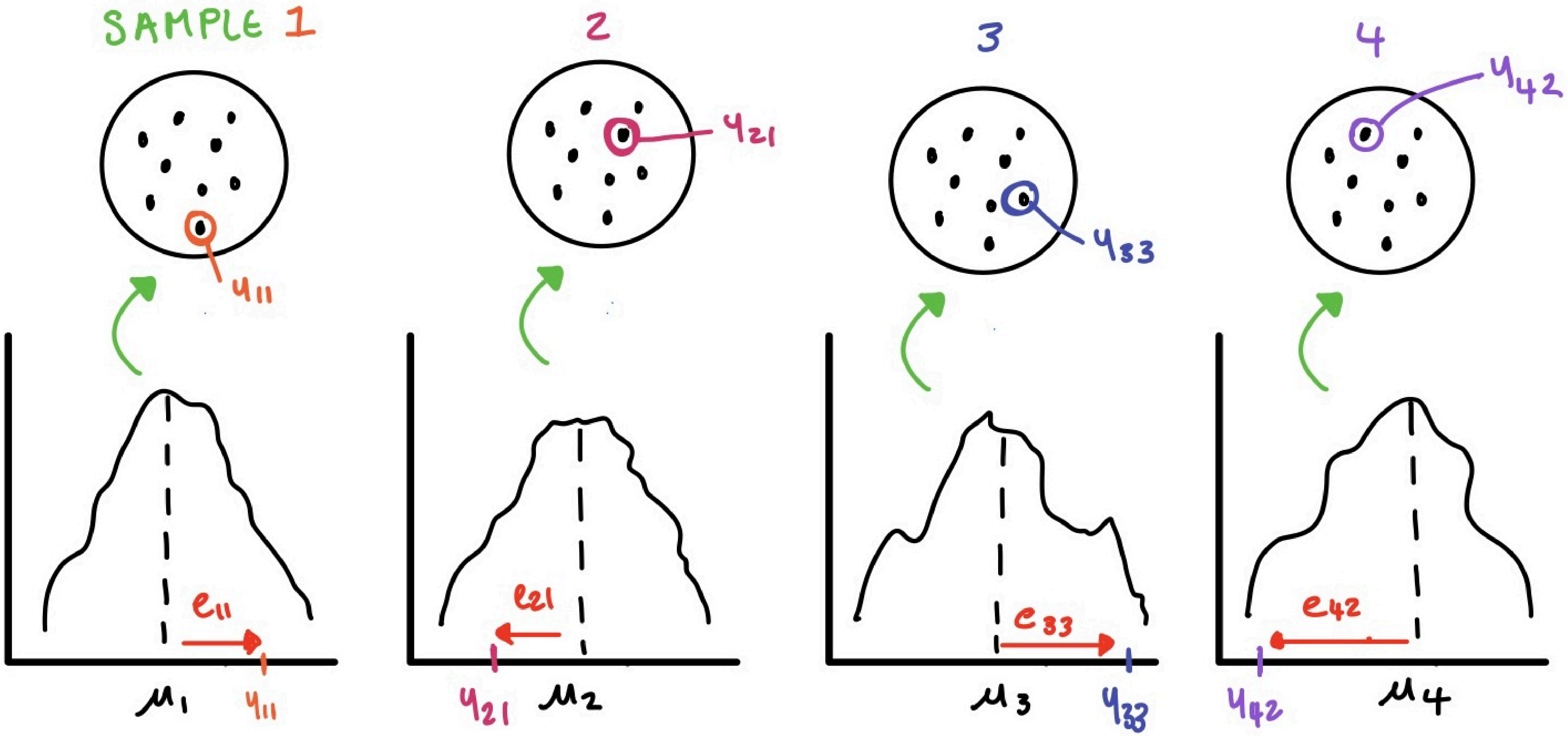
$$Y_{ij} = \mu_i + e_{ij}$$

- This says each observation comes from a group-specific population mean plus some error.
- μ_i is the mean of the population from which a sample is taken and
- Y_{ij} is the j^{th} replicate taken from that sample
- What is the error term?

Visualisation



Visualisation



Reparameterising the model

- We want to test whether the group means are equal:

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_a \quad \text{vs} \quad H_A : \text{At least one } \mu_i \text{ differs}$$

- As is, our model does not naturally lead to this hypothesis test.

So we are going to reparametrise (express our model differently):

- Under H_0: A common baseline (the overall mean)
- Under H_A: Deviations from that mean for each treatment

Visualisation of the model

Reparameterising the model

We write:

$$Y_{ij} = \mu + A_i + e_{ij}$$

Where:

- μ is the grand mean (average across all groups)
- A_i is the effect of treatment i: how much group i differs from the average
- e_i is the usual error

Note it is still the same model!

The grand mean

For this reparameterisation to be valid, we define:

- $\mu = \frac{1}{N} \sum_{i=1}^{N} Y_i$ (the overall mean of all observations) or similarly:
- $\mu = \frac{1}{a} \sum_{i=1}^{a} \mu_i$ (the average of the group means)

and then this means that,

- $\sum_{i=1}^{a} A_i = 0$ (the sum of the treatment effects is zero)

Null hypothesis

We can also rephrase our hypotheses:

$$H0 : \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu$$

- The means are the same.

$$H0 : A_1 = A_2 = A_3 = A_4 = 0$$

- The treatment effects are zero

Alternative hypothesis

$H1\ :\ \text{at least one of the treatment means } (\mu_i) \text{ differ}$

- There is a difference somewhere! Not that all means are different.

$H1\ :\ \text{At least one } A_i\ \neq 0$

- At least one of the treatment effects is not zero.

Why this model?

- Allows us to conduct this global hypothesis test (not obvious why)
- Leads directly to the quantities we need to conduct the hypothesis test via the procedure of ANOVA
- ANOVA partitions the total variation in the response into **different sources**.
- We compare the **variation between groups** to the **variation within groups**.
- If the between-group variation is significantly larger than the within-group variation, we conclude that at least one mean is different.

Do not overcomplicate this!

- We want to compare means
- So we built a model of means
- We want to peform a single test to determine if there are differences between the means
- So we re-express our model i.t.o a common mean and deviations from that mean
- For this parameterisation to be valid and useful, we define:
    - $\mu = \frac{1}{N} \sum_{i=1}^{N} Y_i$ (the overall mean of all observations) which leads to
    - $\sum_{i=1}^{a} A_i = 0$ (the sum of the treatment effects is zero)
- Now with this model we can perform the Analysis of Variance procedure to test the hypothesis

Why do we start with a single global test?

- We are comparing means
- Can you think of a way to do this?

Multiple testing

- When we do a single hypothesis test at 5% significance level: it means 5% of the time (1 out of 20 times) we will reject the hypothesis even though it is true.
- The significance level = Type 1 error rate
- When we conduct many tests, the overall Type 1 Error rate increases
- If we do 20 tests at 5% significance level, the probability of making at least one Type 1 error is:

$$P(\text{at least one Type 1 error}) = 1 - (1 - 0.05)^{20} \approx 0.64$$

What comes after the single test?

- If we reject the null hypothesis in the single test, we know that at least one mean is different.
- But which one(s)?
- We can perform **post-hoc tests** to identify which specific means differ with adjustments for multiple comparisons.

Estimation

Don't know population parameters: μ, A_i, σ^2

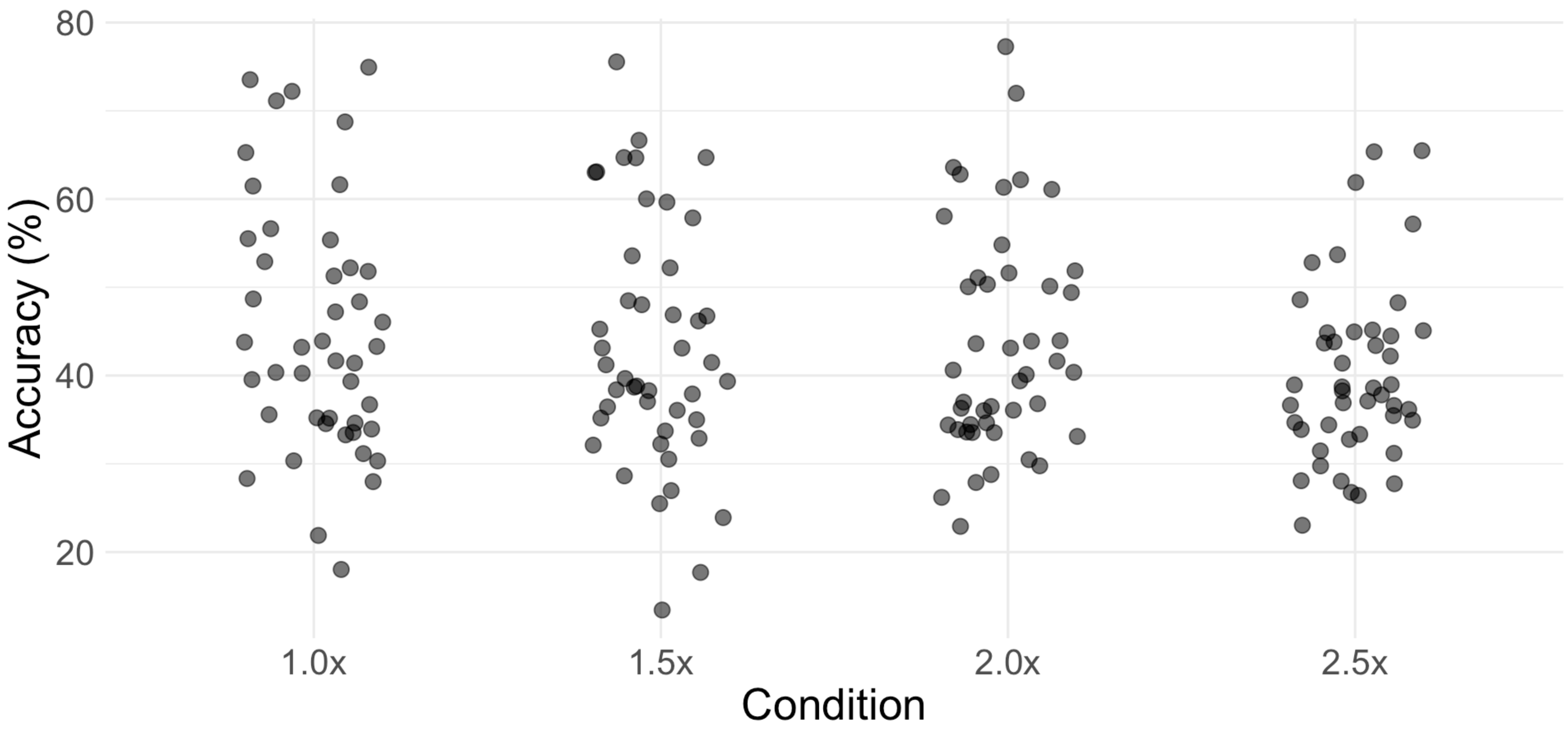
- Find least square estimates which minimise the error sums of squares:

$$\text{SSE} = \sum_i \sum_j e_{ij}^2 = \sum_i \sum_j (Y_{ij} - \hat{Y}_{ij})^2 = \sum_i \sum_j (Y_{ij} - \mu - A_i)^2$$

- Estimate using sample means!

$$\hat{\mu} = \bar{Y}_{..}$$
$$\hat{\mu_i} = \bar{Y}_{i.}$$

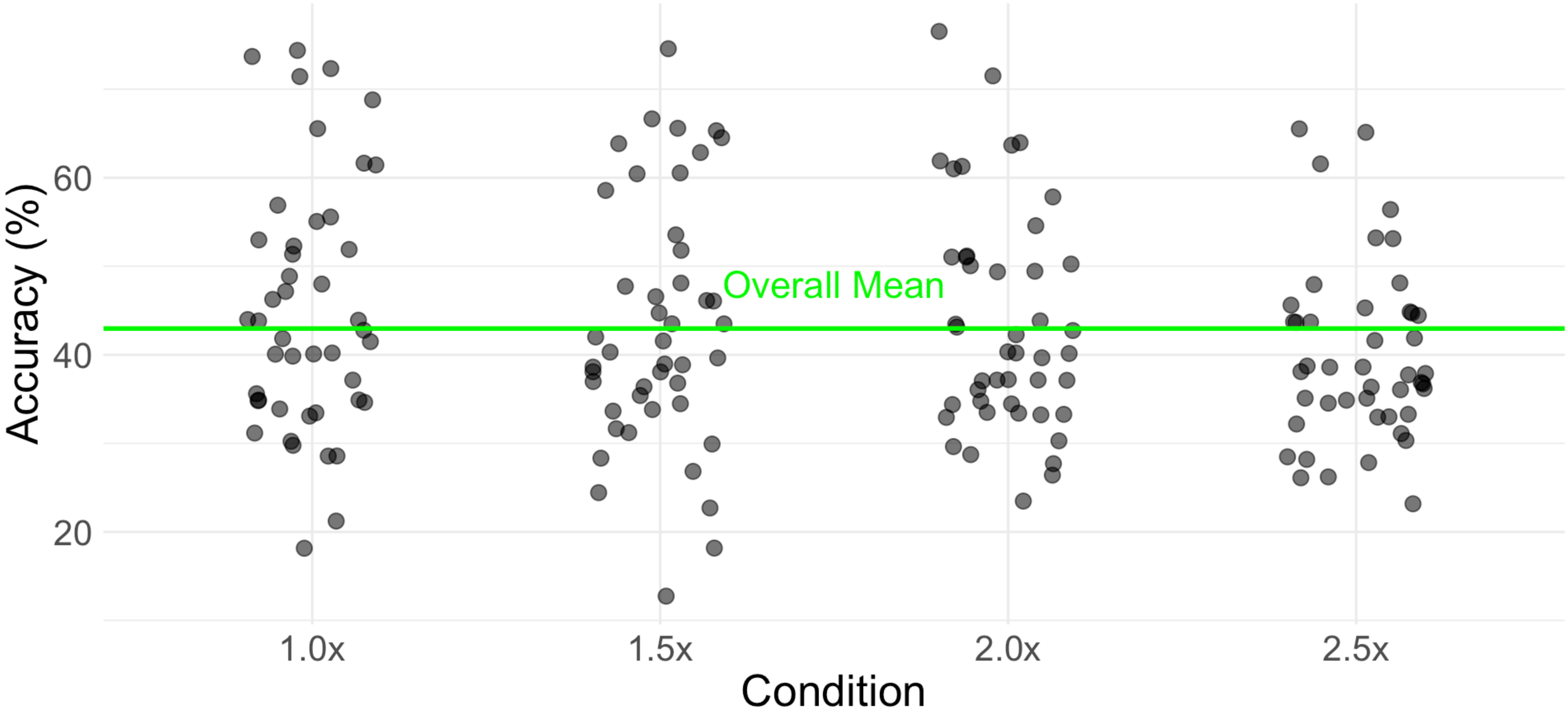
Apply to playback data



What is the estimated overall mean?

```r
overall_mean <- mean(playback$Comprehension)
overall_mean
```

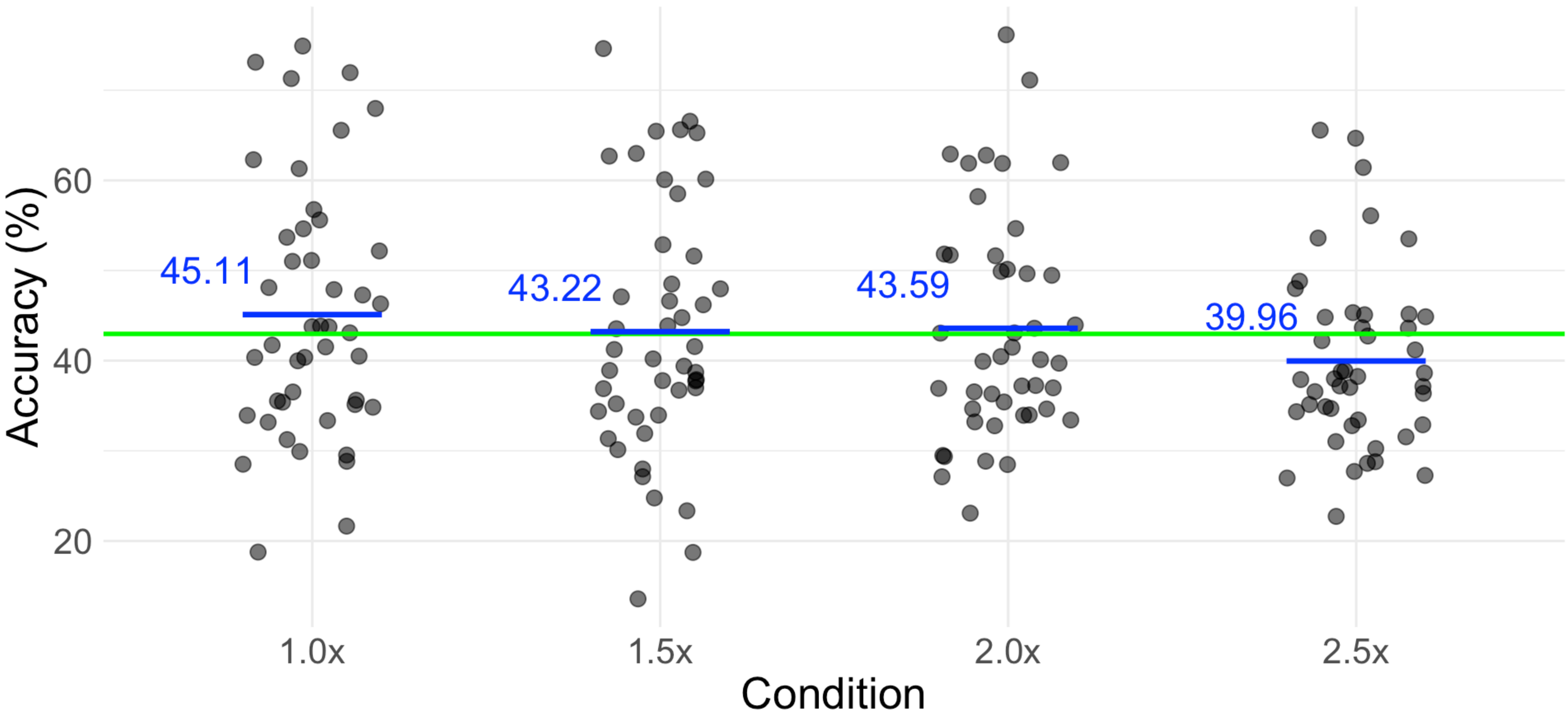
```
[1] 42.972
```


Accuracy (%)
60
40
20
Overall Mean
1.0x
1.5x
2.0x
2.5x
Condition

What are the estimated treatment means?

```r
treatment_means <- tapply(playback$Comprehension, playback$Condition, mean)
treatment_means
```

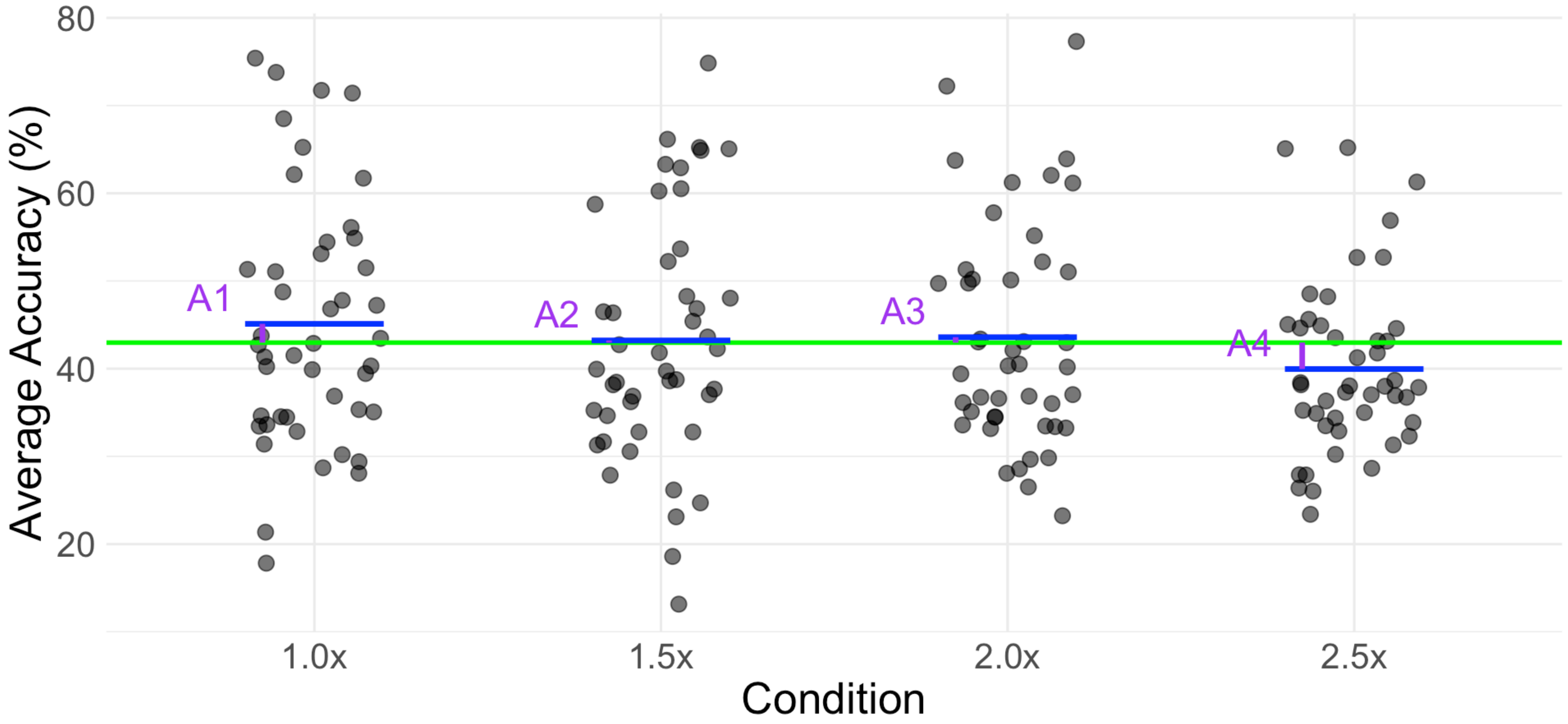
```
    1.0x     1.5x     2.0x     2.5x 
45.11111 43.22178 43.59267 39.96244 
```


Accuracy (%)
60
40
20
45.11
43.22
43.59
39.96
1.0x
1.5x
2.0x
2.5x
Condition

What are the estimated treatment effects?

```
treatment_means - overall_mean
```

```
     1.0x       1.5x       2.0x       2.5x 
 2.1391111  0.2497778  0.6206667 -3.0095556 
```



80
60
40
20
Average Accuracy (%)
A1
A2
A3
A4
1.0x
1.5x
2.0x
2.5x
Condition

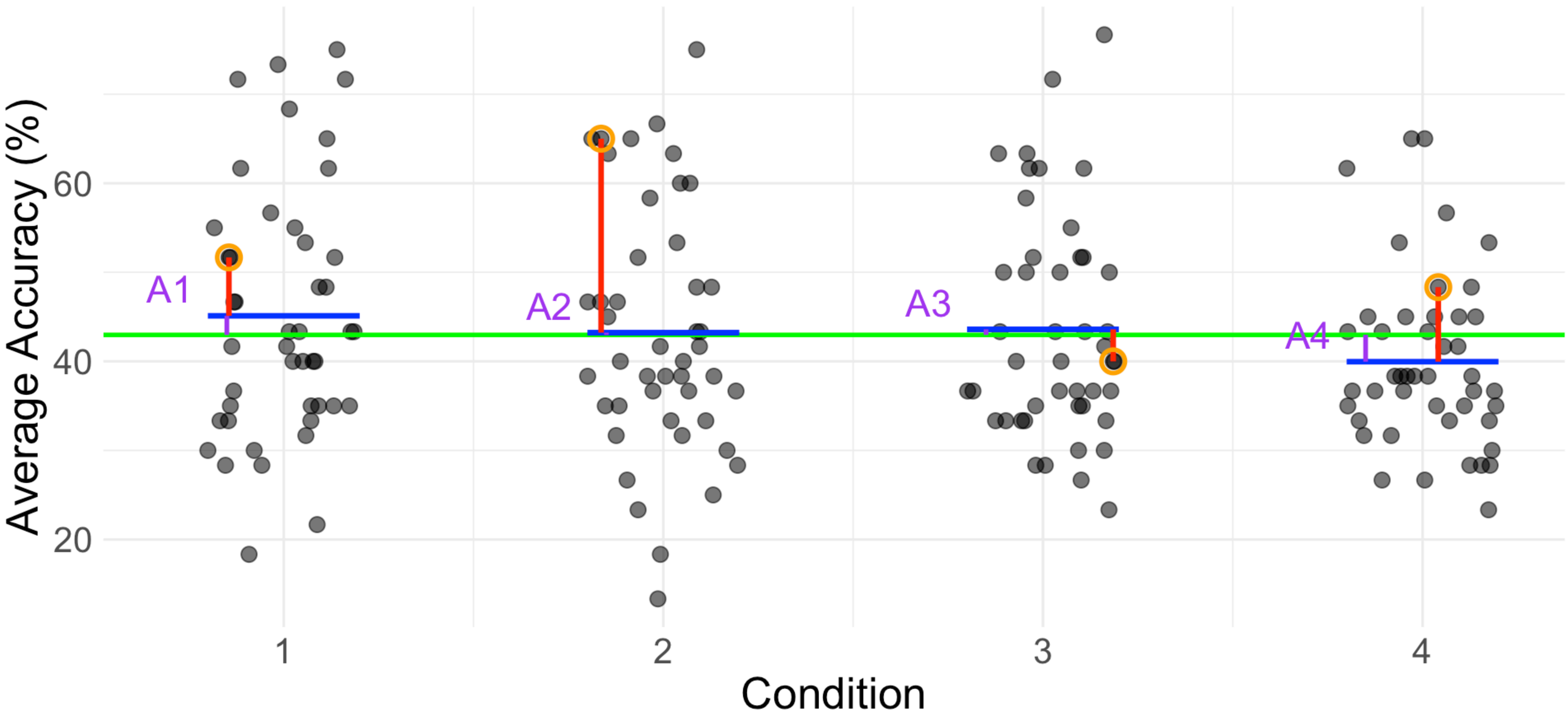
What are the residuals?

$$e_{ij} = Y_{ij} - (\mu + A_i)$$

$$e_{ij} = Y_{ij} - (\mu + (\mu_i - \mu))$$

$$e_{ij} = Y_{ij} - \mu_i$$

$$\hat{e}_{ij} = Y_{ij} - \hat{\mu}_i$$



Average Accuracy (%)
60
40
20
A1
A2
A3
A4
1
2
3
4
Condition

Fitting the model in R

```r
model <- aov(Comprehension ~ Condition, data = playback)
```

Extract model estimates #1

```
model.tables(model, type = "means", se = TRUE)
```

```
Tables of means
Grand mean

42.972

 Condition
Condition
 1.0x  1.5x  2.0x  2.5x
45.11 43.22 43.59 39.96

Standard errors for differences of means
        Condition
            2.687
replic.        45
```

Extract model estimates #2

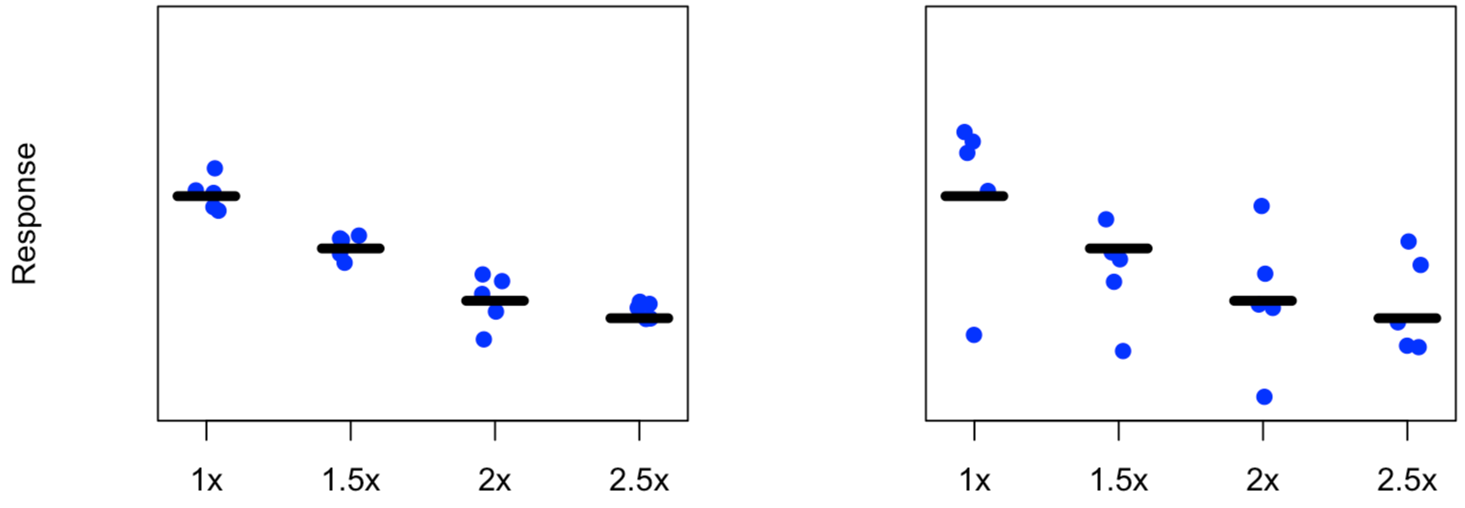
```r
model.tables(model, type = "effects", se = TRUE)
```

```
Tables of effects

 Condition
Condition
   1.0x    1.5x    2.0x    2.5x 
 2.1391  0.2498  0.6207 -3.0096 

Standard errors of effects
        Condition
              1.9
replic.        45
```

Experimental error variance

Let's talk more about $e_{ij} \sim N(0, \sigma^2)$

- Errors = deviations of response values from their group mean.
- Variance of errors = how much the response values deviate from their group mean.



Experimental error variance

- σ^2 is the experimental error variance: how much experimental units deviate within a treatment.
- Constant across groups.
- Estimator:

$$s^2 = \frac{1}{N - a} \sum_{ij} (Y_{ij} - \bar{Y}_{i.})^2$$

- Average of the observed variability across all groups.

But first, model assumptions!

- ANOVA model assumptions:

1. No severe outliers
2. Equal population variance
3. Normally distributed errors
4. Independent errors

- There are formal techniques to test some of these.
- We stick to informal techniques.

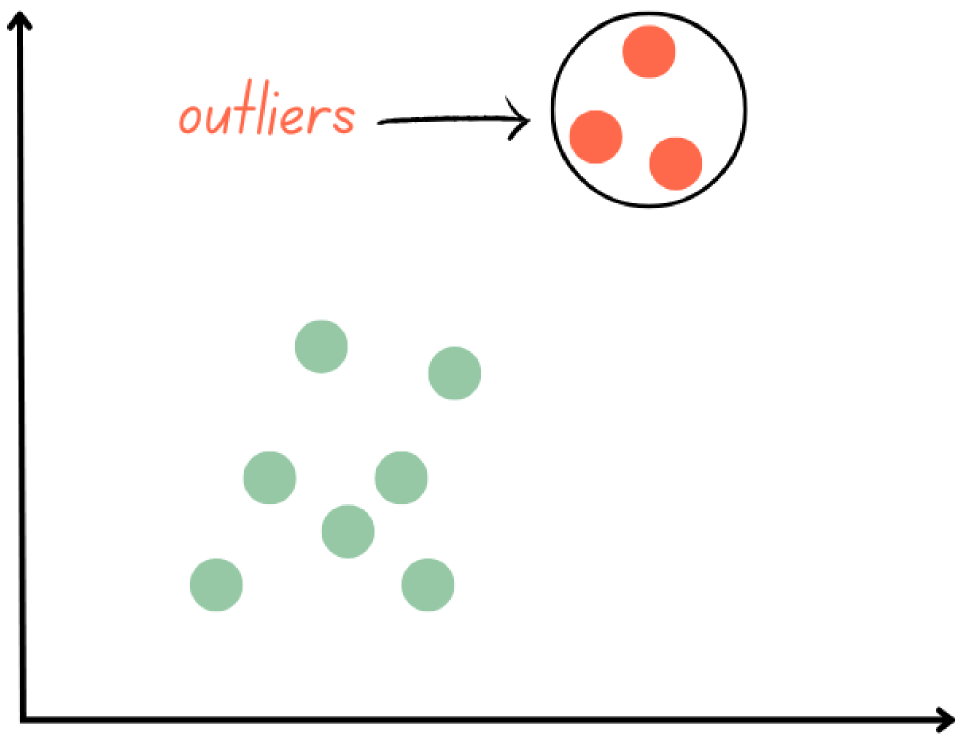
Robustness of ANOVA

- A statistical procedure is said to be robust to departures from a model assumption if the results remain unbiased even when the assumption is not met.

The robustness of ANOVA is as follows:

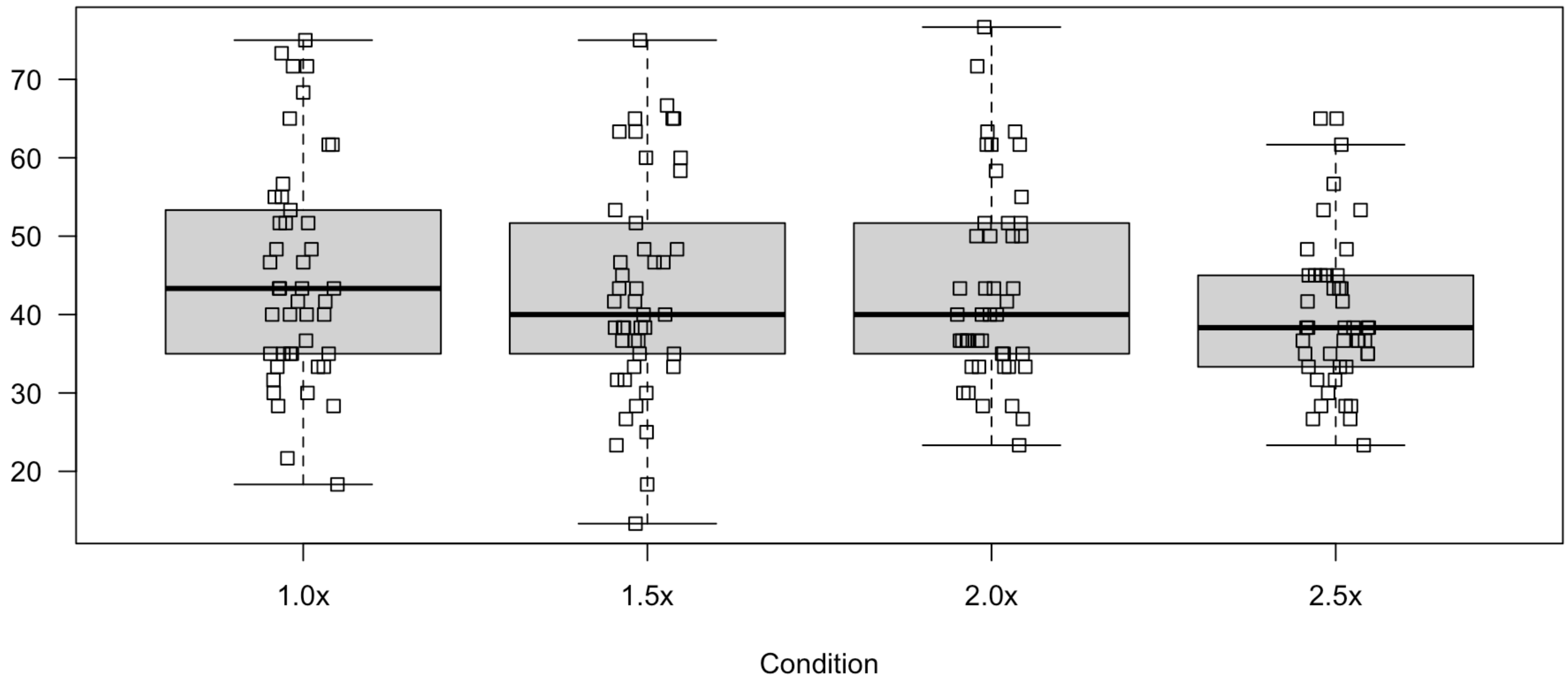
1. Only severe departures from normality such as long-tailed distributions or skewed distributions
2. Independence within and among groups is extremely important.
3. Robust to violations of the equal variance assumption as long as there are no outliers, sample sizes are large and the sample variances are relatively equal.
4. ANOVA is not very resistant to severely outlying observations either.

Outliers



- Data entry or recording errors
- Interesting outliers
- Distort estimates
- Check for outliers by plotting the data

Playback data



Equal population variance

- Assume treatment populations have same variance $e_{ij} \sim N(0, \sigma^2)$
- Simplifies the mathematics and interpretation
- Sample data = sampling variability
- Sample variances need to be similar enough
- Interquartile ranges and standard deviations should be comparable
- Use boxplots again or look at sample statistics.

Equal population variance

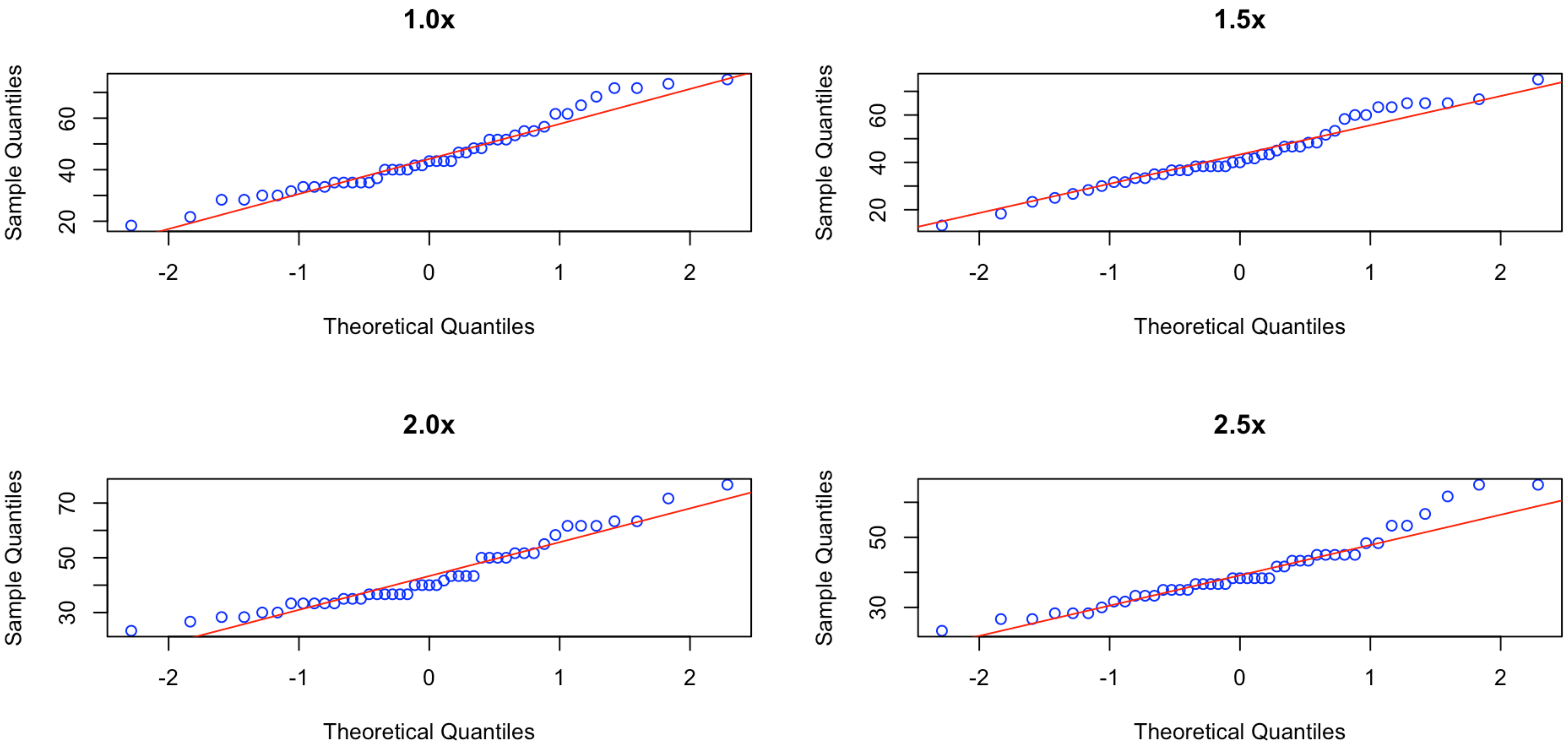
```r
sort(tapply(playback$Comprehension, playback$Condition, sd))
```

```
     2.5x      2.0x      1.5x      1.0x 
 9.818431 12.588622 14.005073 14.095302 
```

Normally distributed errors

- Errors are ND, not raw response!
- But if response is clearly non-normal, our errors will probably also be - why?
- Check with boxplots again or Q-Q plots.
- Look for severe assymmtery.
- Check again after model fitting with residuals!

Q-Q plots


1.0x
1.5x
2.0x
2.5x
Sample Quantiles
Theoretical Quantiles
-2
-1
0
1
2
20
40
60
30
50
70

Independence

- Trickiest assumption of them all.
- This is also about the errors.

Independent errors = determined by independent observations = proper experimental design

- Check after model fitting as well.

Independence

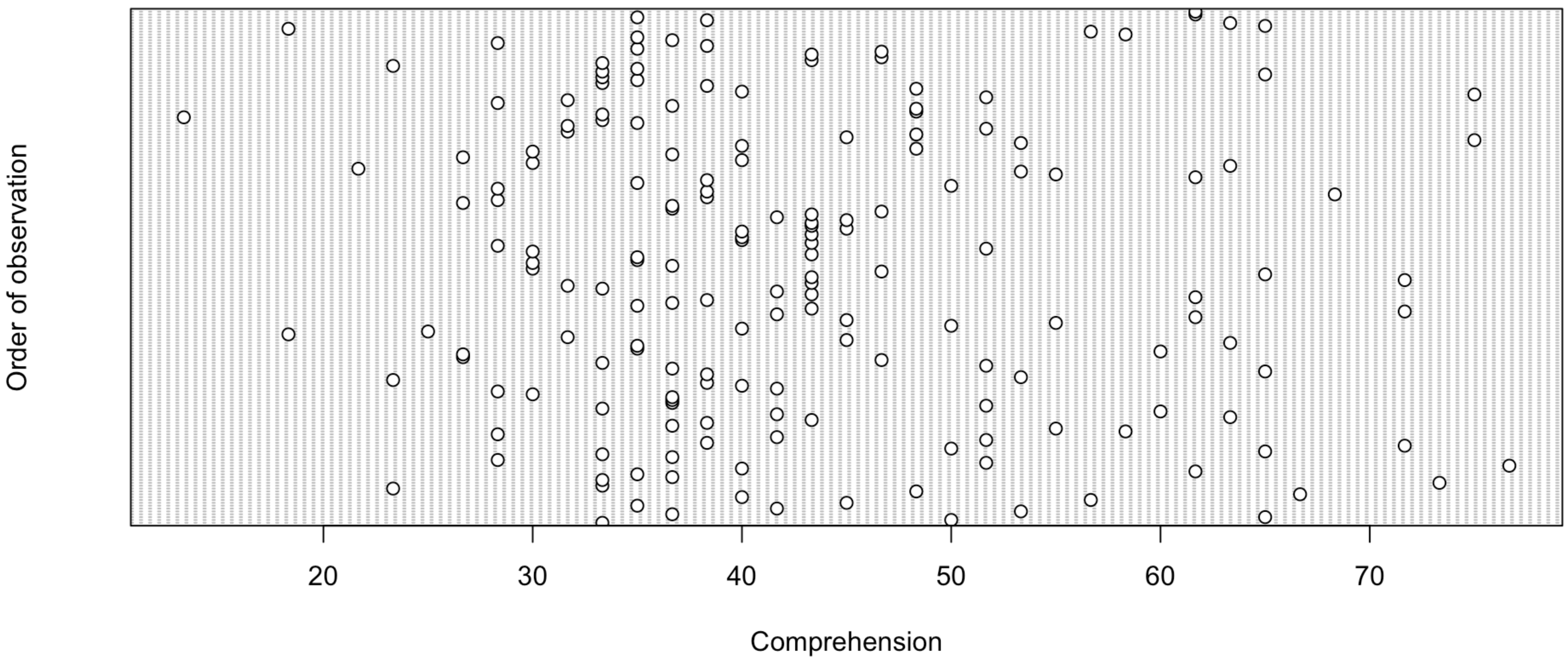
- There should be independence within and among treatments.

Dependence can be caused by:

- Not randomising order in which observations are taken.
- Measurement drift.
- Applying treatments to same group of experimental units.

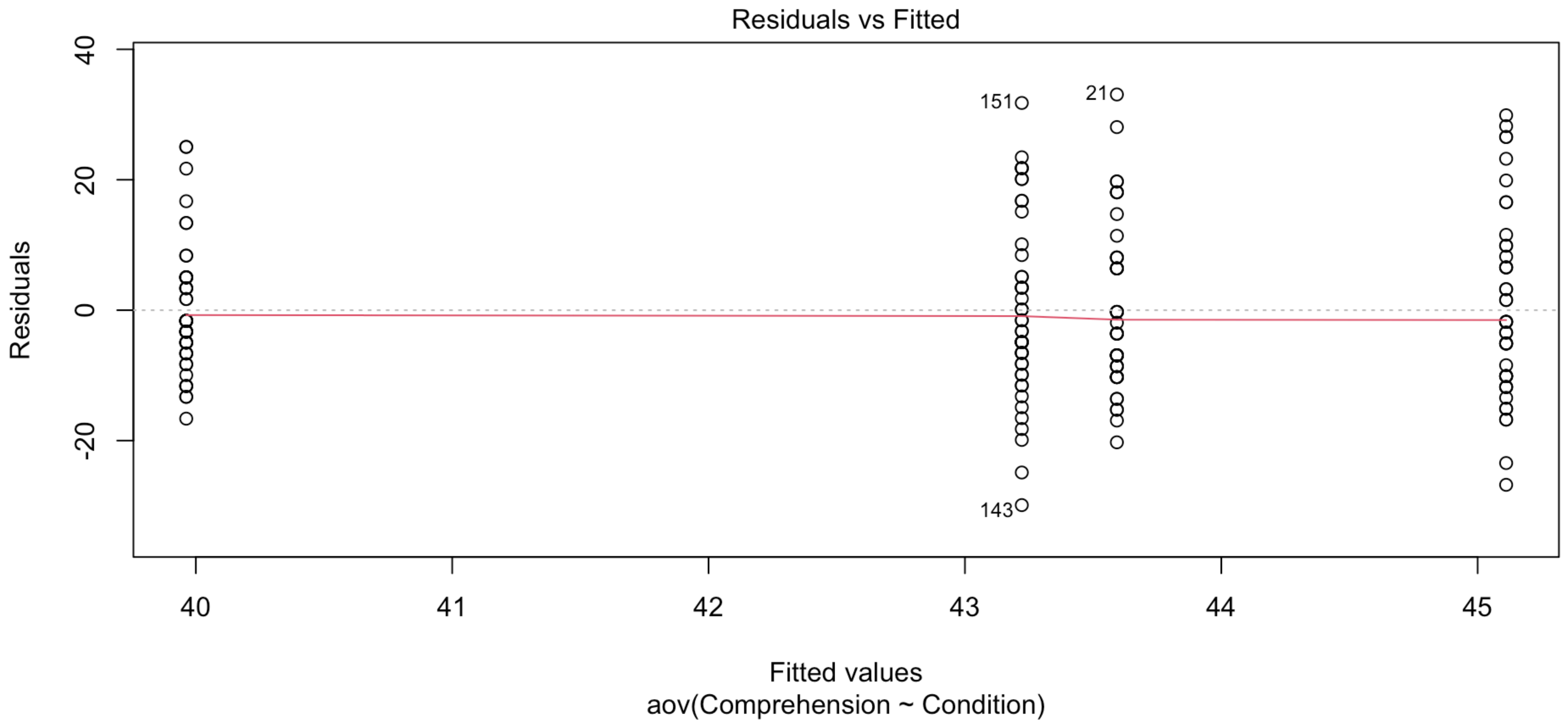
Independence

- Check dependence between successive observations:



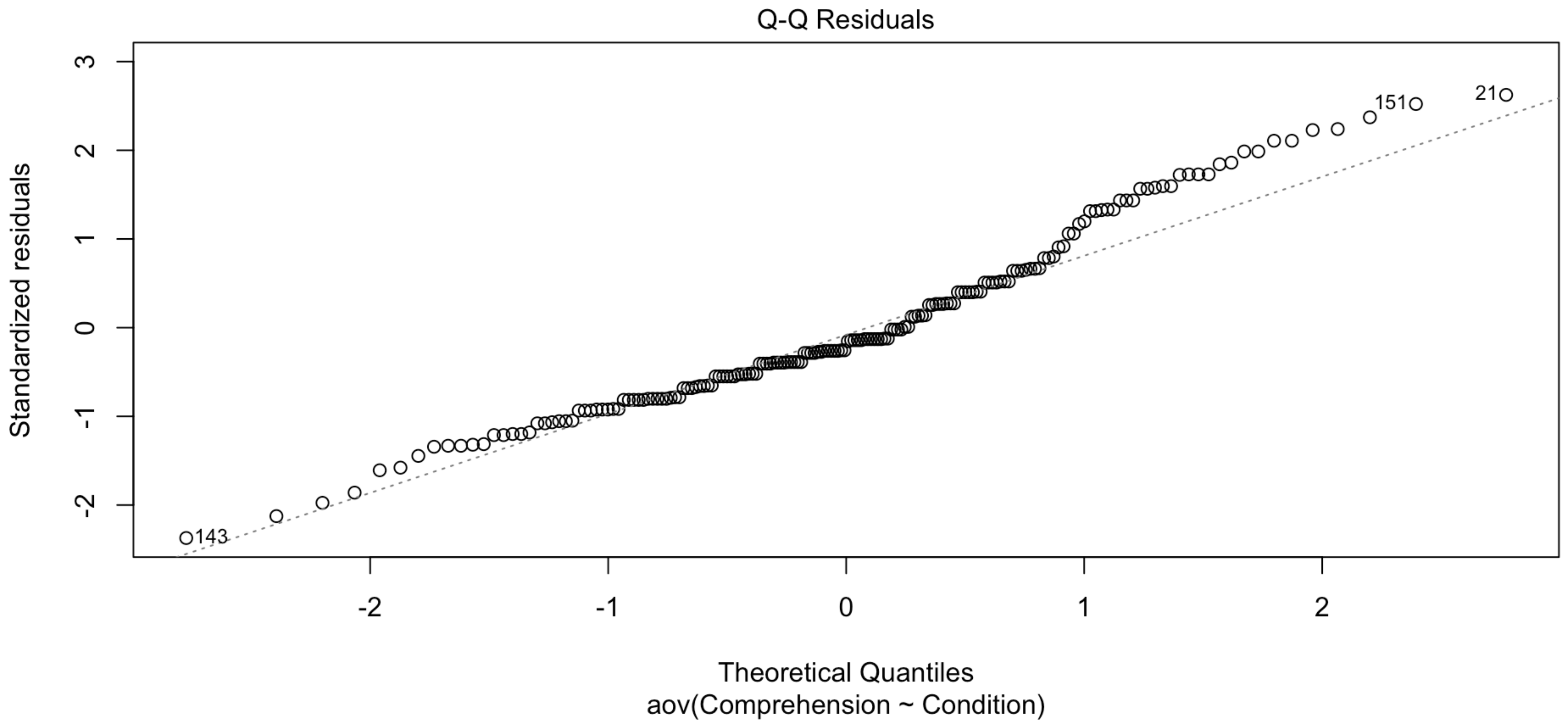
Assumptions after model fitting

```r
plot(model, which = 1)
```



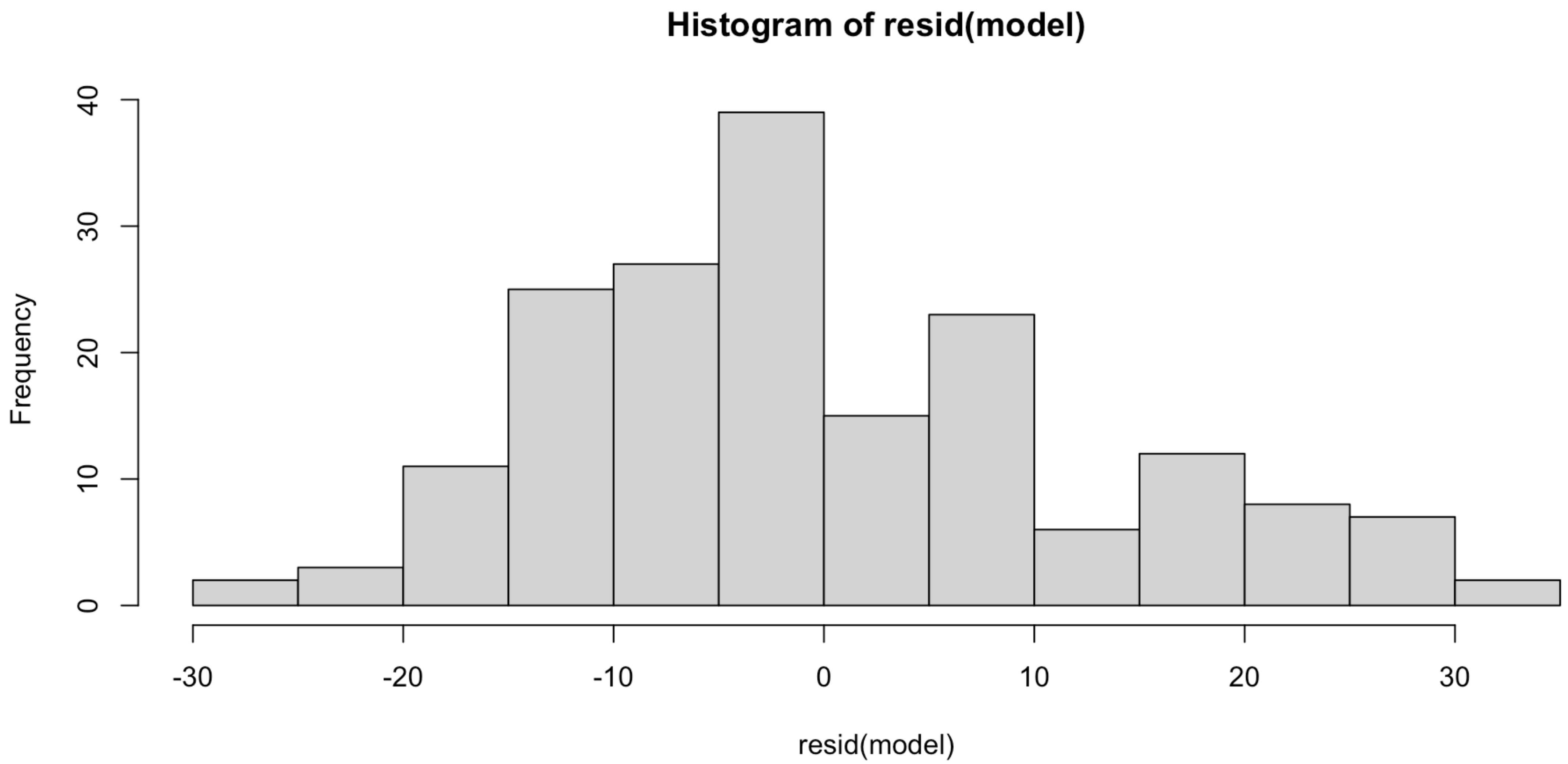
Assumptions after model fitting

```r
plot(model, which = 2)
```



Assumptions after model fitting

```r
hist(resid(model))
```



Assumptions after model fitting

```r
plot(resid(model) ~ seq_along(resid(model)),
     xlab = "Order of Observations",
     ylab = "Residuals",
     main = "Residuals vs. Order")
abline(h = 0, col = "red")
```